Choosing Natural Areas near Orlando, Florida to be Prioritized for Protection from Urban Growth and Development

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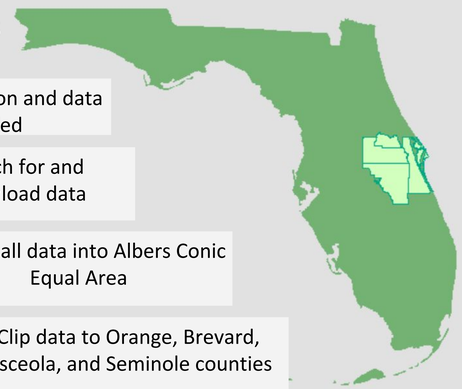
Introduction

Florida has one of the United States' largest wildlife corridors, spanning the entire state from the Everglades to the Panhandle. The connecting corridors between north and south Florida remain threatened by urban growth around Orlando, and are most critical to the continuity of the entire corridor. **This project aims to assess which parts of the Florida wildlife corridor near Orlando should be prioritized for preservation** so that urban growth and a healthy ecosystem can coexist.

First Steps

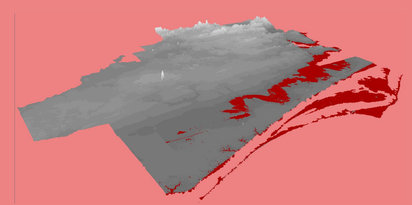


- Define question and data needed
- Search for and download data
- Project all data into Albers Conic Equal Area
- Clip data to Orange, Brevard, Osceola, and Seminole counties
- Define Data as Opportunities or Constraints

Constraint Analyses

Low Elevation

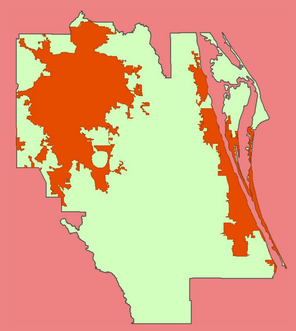
- Mosaic multiple DEM files rasters
- Reclassify raster to above or below 4m
- Convert to polygon



We want to prioritize buying land that will not be underwater with rising sea level (higher elevations.)

Currently Developed

- Existing polygon dataset



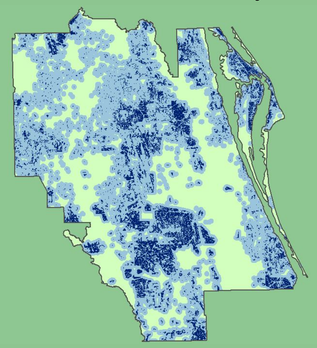
There's no use in buying land that already has development on it!

Models used: Raster-Vector Modeling (Rare Natural Communities, below), 3D modeling in ArcScene (Low Elevation, above), Final Suitability Analysis (right)

Opportunity Analyses

Rare Natural Communities

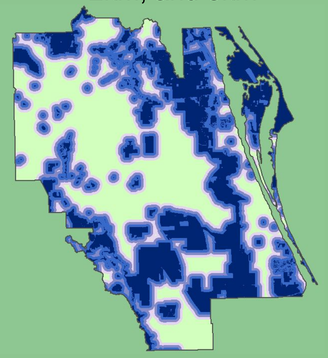
- Raster dataset to polygon dataset
- Buffer rare habitats by 1km



We'd like to add rare habitat types to our network of conserved areas.

Near Existing Conservation

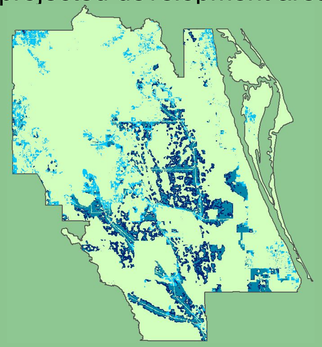
- Buffer existing protected lands by 1km, 2km, and 3km



Land that is near already-conserved areas is more useful for creating wildlife corridors.

Under Threat From Development

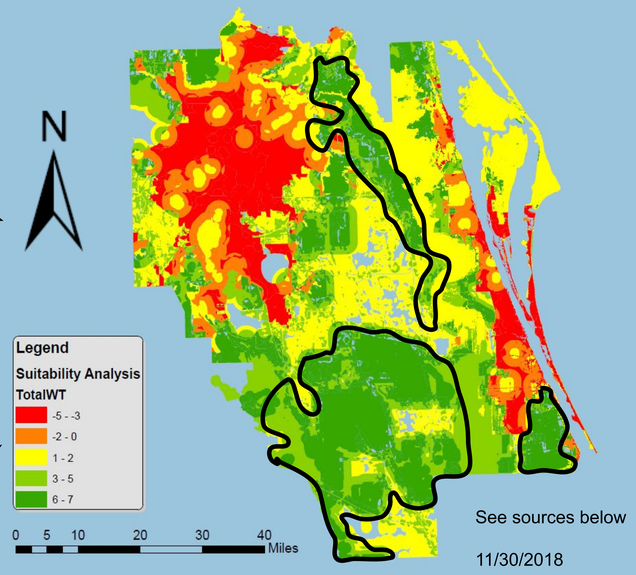
- Erase already-urban areas from projected development areas



Land that is expected to be developed in 2020, 2040, or 2060, but isn't currently urban, is important to protect.

Final Suitability Analysis

- Assign weights to opportunity and constraint factors
- Combine layers and calculate final suitability values



Conclusion

- Several constraint features were not used due to processing limitations (NAIP imagery, interpolating land costs)
- Using those constraints could create more differentiation in suitability values
- Land near Orlando and the Atlantic coast were least optimal due to sea level rise and existing development
- Scale was limited to east-central Florida because statewide datasets were too large
- **Land in the southwest region of the four-county area should be prioritized for purchase by land trusts**

Projection used throughout: Albers Conical Equal Area (Florida Geographic Data Library)
Data sources: Florida Natural Areas Inventory (http://fnai.org/gisdata.cfm), 1000 Friends of Florida (http://www.1000friendsofflorida.org/connecting-people/florida2060), USGS DEM data
Florida corridor information: http://floridawildlifecorridor.org/maps/0